Signal and Image Processing
MILSTONE 3

By:

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GitHub link:

https://github.com/medo-abdelhamid/Image_and_processing_project.git

Milestone 3: CNN Implementation

1. CNN Pipeline and Architecture

For this milestone, I built a Convolutional Neural Network using PyTorch to classify the ball images into the six different categories. To help the model generalize and prevent overfitting, I applied some data augmentation to the training set, including random horizontal flips, 15-degree rotations, and color jitter. All images were resized to 128x128 pixels.

The network is split into two main parts: the convolutional feature extractor and the fully connected classifier.

Convolutional Layers: The feature extractor consists of 6 convolutional blocks. Each block follows the same basic pattern: a 2D Convolution, Batch Normalization, a ReLU activation, and a Max Pooling layer to downsample the spatial dimensions.

- **Block 1:** 3 input channels to 32 filters.
- **Block 2:** 32 filters to 64 filters.
- **Block 3:** 64 filters to 128 filters.
- **Block 4:** 128 filters to 256 filters.
- **Block 5:** 256 filters to 512 filters.
- **Block 6:** 512 filters to 512 filters. Instead of Max Pooling, this final block uses an Adaptive Average Pooling layer `(1, 1)` to ensure the output is perfectly sized for the flat linear layers regardless of slight input variations.

Fully Connected (Dense) Layers: After flattening the output from the conv blocks, the data passes through the classifier section:

- A Linear layer reducing the 512 features down to 128.
- A ReLU activation.
- A Dropout layer set to 0.8. This high dropout rate forces the network to not rely on any single node too heavily.
- A final Linear output layer mapping to our 6 classes (tennis ball, golf ball, football, basketball, baseball, American football).

Training Parameters: The model was trained for 30 epochs using the Adam optimizer (learning rate = 0.001, weight decay = 1e-4). I also used a Cosine Annealing Learning Rate Scheduler to smoothly adjust the learning rate during the training process, alongside a standard Cross-Entropy Loss function.

2. Model Evaluation

The model performed exceptionally well on the test dataset, missing almost zero classifications. The overall metrics are:

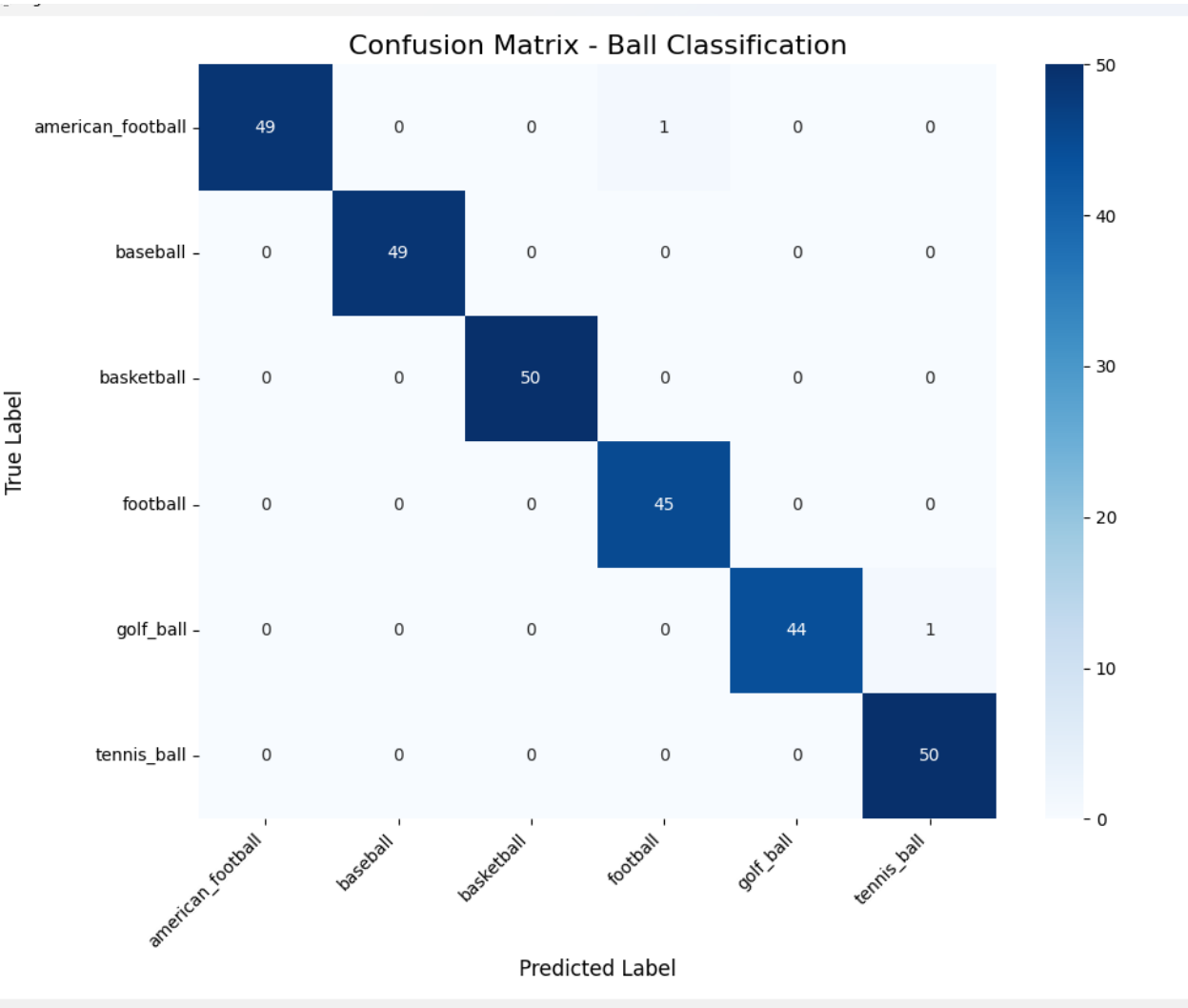
- **Accuracy:** 99.31%
- **Precision (Weighted):** 0.9932
- **Recall (Weighted):** 0.9931

```
Starting Training for 30 epochs...
Epoch 1/30 - Loss: 1.1963
Epoch 2/30 - Loss: 0.7669
Epoch 3/30 - Loss: 0.6819
Epoch 4/30 - Loss: 0.6409
Epoch 5/30 - Loss: 0.5613
Epoch 6/30 - Loss: 0.5676
Epoch 7/30 - Loss: 0.4487
Epoch 8/30 - Loss: 0.4304
Epoch 9/30 - Loss: 0.4719
Epoch 10/30 - Loss: 0.4581
Epoch 11/30 - Loss: 0.4027
Epoch 12/30 - Loss: 0.3302
Epoch 13/30 - Loss: 0.2654
Epoch 14/30 - Loss: 0.2784
Epoch 15/30 - Loss: 0.3035
Epoch 16/30 - Loss: 0.2820
Epoch 17/30 - Loss: 0.2397
Epoch 18/30 - Loss: 0.2517
Epoch 19/30 - Loss: 0.2160
Epoch 20/30 - Loss: 0.1893
Epoch 21/30 - Loss: 0.1986
Epoch 22/30 - Loss: 0.1554
Epoch 23/30 - Loss: 0.1704
Epoch 24/30 - Loss: 0.1378
Epoch 25/30 - Loss: 0.1437
Epoch 26/30 - Loss: 0.1392
Epoch 27/30 - Loss: 0.1344
Epoch 28/30 - Loss: 0.1296
Epoch 29/30 - Loss: 0.1110
Epoch 30/30 - Loss: 0.1232

Total Runtime for Training: 14.59 minutes

Starting Evaluation...
Accuracy: 99.31%
Precision (Weighted): 0.9932
Recall (Weighted): 0.9931
Confusion Matrix:
[[49  0  0  1  0  0]
 [ 0 49  0  0  0  0]
 [ 0  0 50  0  0  0]
 [ 0  0  0 45  0  0]
 [ 0  0  0  0 44  1]
 [ 0  0  0  0  0 50]]
```

Confusion Matrix



3. Total Runtime

The model was trained over 30 epochs, and the total runtime for training was **14.59 minutes**.

4. Top 3 Probabilities

When scanning the entire test set to find the predictions with the highest confidence, the model was 100% certain about these top three examples:

1. **Predicted:** baseball (100.0000%) | **True Label:** baseball
2. **Predicted:** football (100.0000%) | **True Label:** football
3. **Predicted:** baseball (100.0000%) | **True Label:** baseball

```
Scanning the entire test set for the top 3 best predictions

 TOP 3
1. Predicted: baseball (100.0000%) | True Label: baseball
2. Predicted: football (100.0000%) | True Label: football
3. Predicted: baseball (100.0000%) | True Label: baseball
```

Milestone 2: Previous Implementation

1. Updated Pipeline

In Milestone 2, the pipeline relied on traditional computer vision techniques rather than deep learning. During optimization, I found that an image resize of 128x128 lost too much detail, so the resolution was increased to 256x256, which significantly improved the clarity of texture features.

For the classification model, I initially experimented with SVM and k-NN, but their accuracy was quite low. Because the combined feature vector was very large and complex, I ultimately updated the pipeline to use a Random Forest classifier with 500 trees, which handled the complexity much better and yielded a final accuracy of 95.85%.

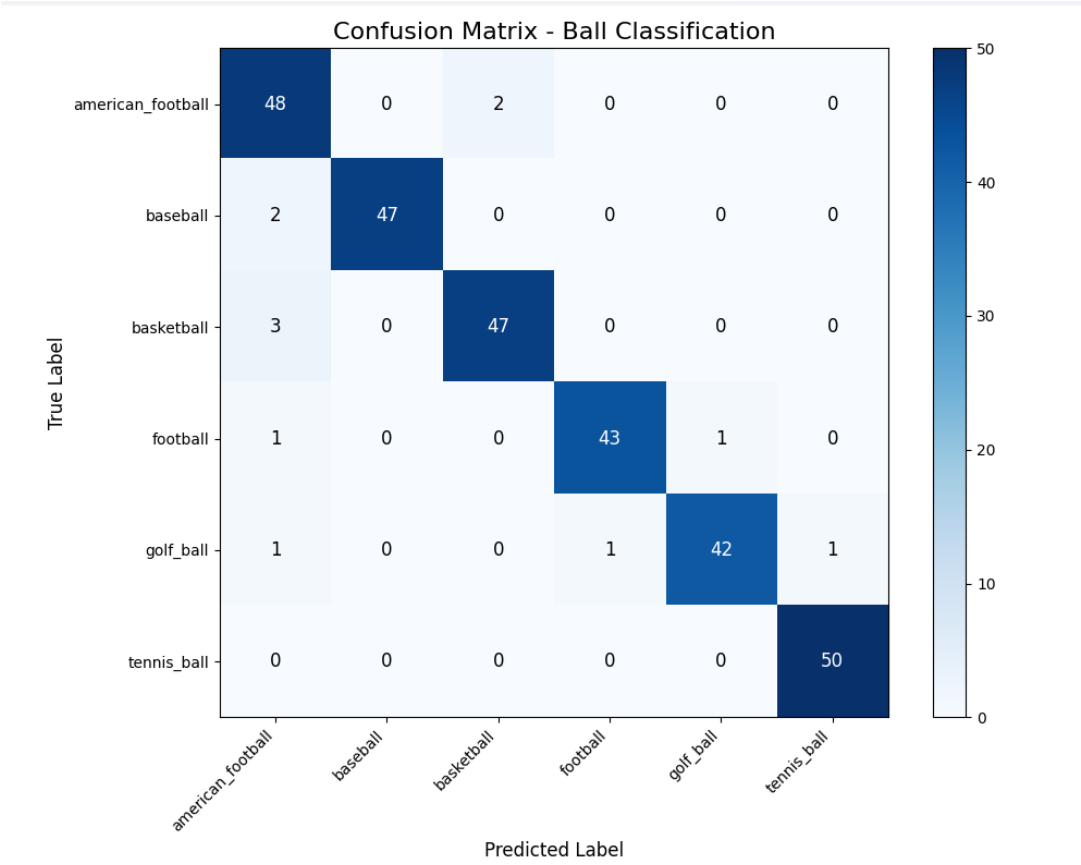
2. Output Feature Vector

Instead of the CNN automatically learning features, I manually engineered a comprehensive feature vector to capture every detail of the balls. The final output feature vector is a combination of the following extractions:

- **Color Histograms:** 3D RGB and 1D grayscale histograms to capture primary colors.
- **GLCM (Gray-Level Co-occurrence Matrix):** To identify surface textures through contrast, dissimilarity, homogeneity, energy, and correlation.
- **HOG (Histograms of Oriented Gradients):** To capture overall structure and directional edges.
- **SIFT (Scale Invariant Feature Transform):** Tuned specifically to 65 keypoints to find unique features regardless of rotation.
- **Contour Features:** Calculating the number of contours, area, and circularity.
- **DoG (Difference of Gaussians):** Used to highlight fine edges and missed details.
- **K-Means Features:** Clustering pixel values to find dominant colors and their proportions.
- **Hough Circle Transform:** Specifically added to detect the radius and center, giving the model a hint about the circular shape and size.

3. Confusion Matrix

Here is the confusion matrix for the Random Forest model from Milestone 2, which achieved an overall accuracy of 95.85%



4. Total Runtime

The whole process took somewhere between 3 to 5 minutes to complete.